

Prim's MST

Minimum Spanning Trees, Prim's Algorithm · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★ Advanced

Problem

Compute a minimum spanning tree with Prim's algorithm, growing one tree from vertex 0 using a min-heap. Report the total weight and the order vertices are added.

Implement the heap-based growth from scratch.

What to Compute

- **MST Weight:** the total weight of the spanning tree grown from vertex 0.
- **Build Order:** the order vertices are added to the tree.

Input / Output Format

Input: Line 1: N M. Next M lines: an undirected weighted edge u v w.

Output: Two lines: MST Weight, Build Order.

Examples

Example 1

Input: 4 5 0 1 1 0 2 3 1 2 1 1 3 4 2 3 2

Output:

MST Weight: 4

Build Order: 0 1 2 3

Explanation: From 0 the cheapest reach is 1 (weight 1), then 2 (weight 1), then 3 (weight 2), total 4.

Example 2

Input: 3 3 0 1 1 1 2 2 0 2 3

Output:

MST Weight: 3

Build Order: 0 1 2

Explanation: From 0 take 0-1 (1) then 1-2 (2); total 3, vertices added 0, 1, 2.

Constraints

- Start from vertex 0.
- Always extend by the cheapest edge to an unvisited vertex.
- Keep adjacency lists sorted for a deterministic build order.

Sample Run

Sample Input

```
4 5
0 1 1
0 2 3
1 2 1
1 3 4
2 3 2
```

Sample Output

```
MST Weight: 4
Build Order: 0 1 2 3
```